Intelligent Data Analytics Platform with AI-Powered Insights

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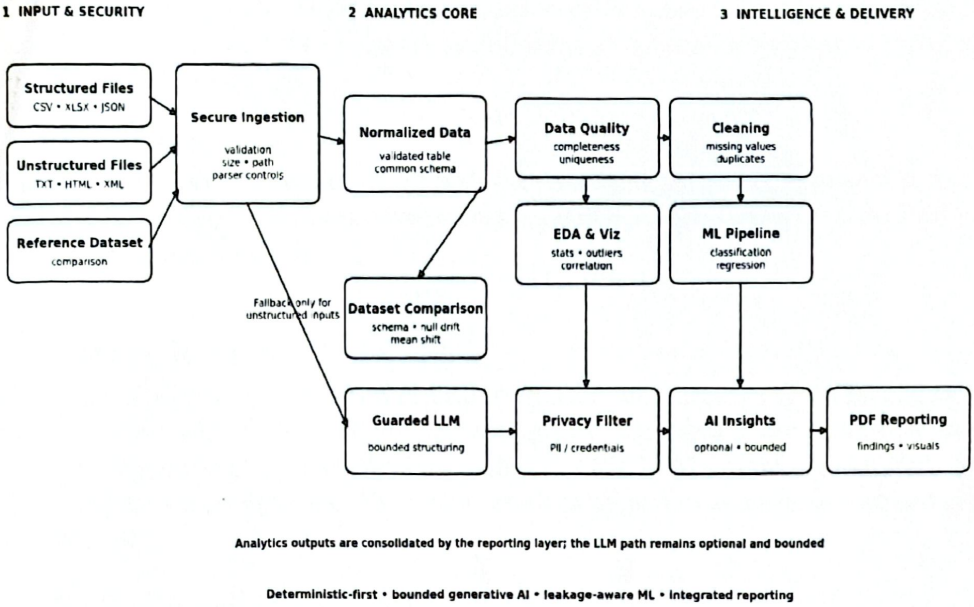
1. Abstract

Modern data analytics workflows commonly require separate stages for data ingestion, preparation, exploratory analysis, predictive modeling, and reporting. This fragmentation can increase operational effort and complicate reproducibility and privacy controls when unstructured information is processed with large language models (LLMs). Intelligent Data Analytics Platform with AI-Powered Insights is a local-first intelligent analytics platform designed to integrate these stages within a unified workflow. The system combines deterministic multi-format data ingestion with a bounded LLM fallback for unstructured tabular extraction. It evaluates data reliability using completeness and uniqueness measures, supports schema and selected distributional comparisons, and provides automated cleaning, exploratory data analysis, interactive visualization, and supervised machine learning through leakage-aware scikit-learn pipelines. A privacy layer screens potentially sensitive information and credential-like patterns before optional LLM transmission, while an executive reporting layer combines analytical summaries, visualizations, quality indicators, and AI-assisted insights into a structured PDF report. The platform therefore provides an integrated bridge between deterministic data engineering and guarded generative intelligence. Its principal contribution is an architectural and systems-integration approach that emphasizes deterministic processing wherever possible and restricts generative processing to bounded tasks requiring unstructured-data interpretation.

Keywords: data analytics; data quality; exploratory data analysis; machine learning; large language models; privacy; dataset shift; automated reporting.

2. Proposed Method

The proposed method is a layered, deterministic-first analytics architecture. Inputs are first validated and parsed through controlled ingestion. The analytics orchestrator then routes the normalized dataset through quality, cleaning, exploratory, comparison, and machine-learning services. The guarded LLM gateway remains a bounded alternative for unstructured inputs, while the privacy layer acts as a cross-cutting boundary before optional external AI processing. Results are consolidated by the reporting layer.



2.1 Secure Multi-Format Ingestion

The ingestion layer accepts the implemented formats—CSV, TSV, XLSX, JSON, JSONL, Parquet, XML, HTML, and TXT—through deterministic parsers. Upload-size controls, path-containment checks, and XML DTD/entity rejection provide input-level safeguards before analysis.

2.2 Guarded LLM Structuring

If deterministic parsing is insufficient, unstructured input may be routed through a bounded Gemini-based structuring path. Imported content is treated as untrusted data; prompt content is delimited; and returned tabular content is subjected to bounded extraction limits on characters, lines, candidates, and columns.

2.3 Analytics Orchestration and Data Quality

The orchestration layer coordinates the normalized dataset through quality assessment and downstream analytics. Completeness and uniqueness are calculated and combined into the Quality Index. Dataset comparison checks schema differences, missingness changes, and relative numeric mean shifts.

2.4 Cleaning, EDA, and Visualization

Cleaning supports duplicate removal and configurable missing-value treatment using mean, median, zero, mode, constant, forward-fill, and backward-fill strategies. EDA provides descriptive statistics, cardinality and missingness summaries, Tukey IQR-based outlier analysis, Pearson correlations, and interactive visualizations.

2.5 Leakage-Aware Machine Learning

The system identifies classification or regression workflows and uses scikit-learn Pipeline and ColumnTransformer boundaries for imputation, scaling, and categorical encoding. This keeps preprocessing within the model workflow and reduces the risk of information from evaluation data influencing training transformations.

2.6 Privacy Boundary and AI Insights

The privacy layer uses heuristics for emails, provider tokens, phone numbers, IBANs, and payment-card patterns, including Luhn validation for card numbers. Sensitive fields can be excluded from LLM context. AI-generated insights are therefore positioned as an optional bounded layer rather than the authoritative computational path.

2.7 Executive Reporting

The reporting layer consolidates analytical findings, visualizations, quality indicators, and optional AI-assisted strategic insights into a structured multi-section PDF report. The report is the final presentation layer for the analytics workflow.

3. Literature Review

The literature is organized around the technical components that directly support Intelligent Data Analytics Platform with AI-Powered Insights: data quality, data preparation, exploratory analysis, machine learning, leakage and reproducibility, tabular LLMs, LLM security, dataset shift, visual analytics, and automated reporting. The review positions the project as an integration and systems-architecture effort.

3.1 Data Quality and Data Preparation

Wang and Strong [1] establish a multidimensional view of data quality beyond accuracy alone, while Batini et al. [2] review methodologies for assessing and improving data quality. Kandel et al. [3] demonstrate interactive data transformation, and Wickham [11] formalizes principles for organizing data for analysis. These works support Intelligent Data Analytics Platform with AI-Powered Insights's explicit quality and cleaning stages.

3.2 Exploratory Analysis and Visualization

Tukey [4] provides the foundation for exploratory data analysis through statistical and graphical investigation. Wilkinson [18] frames visualization as a grammar for representing data, while Satyanarayan et al. [19] describe declarative visualization through Vega-Lite. These works support the project's descriptive, outlier, correlation, and interactive visualization workflow.

3.3 Machine Learning and Leakage Control

Pedregosa et al. [5] provide the scikit-learn foundation used by the project. Breiman [13], Friedman [14], and Chen and Guestrin [15] represent established supervised learning approaches relevant to the model layer. Kapoor and Narayanan [6] and Kaufman et al. [12] emphasize the methodological risks associated with leakage and evaluation practices, supporting Intelligent Data Analytics Platform with AI-Powered Insights's use of Pipeline and ColumnTransformer boundaries.

3.4 LLMs for Tabular Data and Data Preparation

Fang et al. [7] survey LLM applications to tabular data, covering prediction, generation, question answering, and table understanding. Akella et al. [20] examine LLM-supported data-wrangling

automation, while Chen et al. [21] survey language models for tabular data preparation. These works provide direct context for Intelligent Data Analytics Platform with AI-Powered Insights's bounded LLM structuring and AI-assisted analytics functions.

3.5 LLM Security and Privacy

Greshake et al. [8] demonstrate indirect prompt-injection risks in LLM-integrated applications. This supports treating imported content as untrusted and isolating instructions from data. The project's privacy layer consequently acts as a defense-in-depth boundary before optional external LLM processing.

3.6 Dataset Shift and Reliability

Rabanser et al. [9] investigate methods for detecting dataset shift. Intelligent Data Analytics Platform with AI-Powered Insights adopts a narrower batch-comparison approach based on schema differences, missingness changes, and numeric mean shifts. The implementation should therefore be described as lightweight dataset auditing rather than comprehensive concept-drift detection.

3.7 Automated Reporting and Data-to-Document Generation

Wiseman et al. [10] study the challenges of generating documents from structured data. Intelligent Data Analytics Platform with AI-Powered Insights applies this principle at the systems level by assembling analytical results and visualizations into structured executive PDF reports rather than proposing a new data-to-text generation model.

3.8 Research Gap and Project Contribution

Existing work addresses individual components of the analytics workflow, including data quality, transformation, visualization, machine learning, dataset-shift detection, tabular language-model applications, and LLM security. Intelligent Data Analytics Platform with AI-Powered Insights addresses the integration of these components into one deterministic-first workflow with a bounded generative-AI path, explicit privacy screening, leakage-aware machine learning, dataset comparison, and automated reporting. The defensible contribution is therefore an architectural and systems-integration contribution rather than a claim that the underlying algorithms are individually novel.

4. References

[1] R. Y. Wang and D. M. Strong, "Beyond Accuracy: What Data Quality Means to Data Consumers," Journal of Management Information Systems, vol. 12, no. 4, pp. 5–33, 1996, doi: 10.1080/07421222.1996.11518099.

[2] C. Batini, C. Cappiello, C. Francalanci, and A. Maurino, "Methodologies for Data Quality Assessment and Improvement," ACM Computing Surveys, vol. 41, no. 3, 2009, doi: 10.1145/1541880.1541883.

[3] S. Kandel, A. Paepcke, J. M. Hellerstein, and J. Heer, "Wrangler: Interactive Visual Specification of Data Transformation Scripts," in Proc. ACM CHI, 2011, pp. 3363–3372, doi: 10.1145/1978942.1979444.

[4] J. W. Tukey, Exploratory Data Analysis. Reading, MA, USA: Addison-Wesley, 1977.

[5] F. Pedregosa et al., "Scikit-learn: Machine Learning in Python," Journal of Machine Learning Research, vol. 12, pp. 2825–2830, 2011.

[6] S. Kapoor and A. Narayanan, "Leakage and the Reproducibility Crisis in Machine-Learning-Based Science," Patterns, vol. 4, no. 9, 100804, 2023, doi: 10.1016/j.patter.2023.100804.

[7] X. Fang, W. Xu, F. A. Tan, Z. Hu, J. Zhang, Y. Qi, S. H. Sengamedu, and C. Faloutsos, "Large Language Models (LLMs) on Tabular Data: Prediction, Generation, and Understanding—A Survey," Transactions on Machine Learning Research, 2024, arXiv:2402.17944.

[8] K. Greshake, S. Abdelnabi, S. Mishra, C. Endres, T. Holz, and M. Fritz, "Not What You've Signed Up For: Compromising Real-World LLM-Integrated Applications with Indirect Prompt Injection," in Proc. ACM Workshop on Artificial Intelligence and Security, 2023, pp. 79–90, doi: 10.1145/3605764.3623985.

[9] S. Rabanser, S. Günnemann, and Z. C. Lipton, "Failing Loudly: An Empirical Study of Methods for Detecting Dataset Shift," in Advances in Neural Information Processing Systems, vol. 32, 2019, pp. 1394–1406.

[10] S. Wiseman, S. M. Shieber, and A. M. Rush, "Challenges in Data-to-Document Generation," in Proc. 2017 Conference on Empirical Methods in Natural Language Processing, 2017, pp. 2253–2263, doi: 10.18653/v1/D17-1239.

[11] H. Wickham, "Tidy Data," Journal of Statistical Software, vol. 59, no. 10, pp. 1–23, 2014, doi: 10.18637/jss.v059.i10.

[12] S. Kaufman, S. Rosset, C. Perlich, and O. Stitelman, "Leakage in Data Mining: Formulation, Detection, and Avoidance," ACM Transactions on Knowledge Discovery from Data, vol. 6, no. 4, 2012, doi: 10.1145/2382577.2382579.

[13] L. Breiman, "Random Forests," Machine Learning, vol. 45, pp. 5–32, 2001, doi: 10.1023/A:1010933404324.

[14] J. H. Friedman, "Greedy Function Approximation: A Gradient Boosting Machine," Annals of Statistics, vol. 29, no. 5, pp. 1189–1232, 2001, doi: 10.1214/aos/1013203451.

[15] T. Chen and C. Guestrin, "XGBoost: A Scalable Tree Boosting System," in Proc. 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, 2016, pp. 785–794, doi: 10.1145/2939672.2939785.

[16] R. J. Little and D. B. Rubin, Statistical Analysis with Missing Data, 2nd ed. Hoboken, NJ, USA: Wiley, 2002.

[17] W. McKinney, "Data Structures for Statistical Computing in Python," in Proc. 9th Python in Science Conference, 2010, pp. 56–61, doi: 10.25080/Majora-92bf1922-00a.

[18] L. Wilkinson, The Grammar of Graphics, 2nd ed. New York, NY, USA: Springer, 2005.

[19] A. Satyanarayan, R. Russell, J. Hoffswell, and J. Heer, "Reactive Vega: A Streaming Dataflow Architecture for Declarative Interactive Visualization," IEEE Transactions on Visualization and Computer Graphics, vol. 22, no. 1, pp. 659–668, 2016, doi: 10.1109/TVCG.2015.2467091.

[20] A. Akella, A. Manatkar, K. Narayanam, and S. Mehta, "CodeGenWrangler: Data Wrangling Task Automation Using Code-Generating Models," in Proc. NAACL-HLT 2025 Industry Track, pp. 949–960, 2025, doi: 10.18653/v1/2025.naacl-industry.70.

[21] M. Chen, Y. Sun, T. Li, J. Wang, K. Wang, X. Lin, Y. Zhang, and W. Zhang, "Empowering Tabular Data Preparation with Language Models: Why and How?," arXiv preprint arXiv:2508.01556, 2025, doi: 10.48550/arXiv.2508.01556.